Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: 20260206_171419_resnet50_classifier.pt

Attack Type: FGSM
Images Evaluated: 10

Attack Success Rate	Original Confidence	Adversarial Confidence	Execution Time
100.0%	34.13%	34.70%	8.78s
			Accuracy drop: -0.57%

Threat Level Assessment

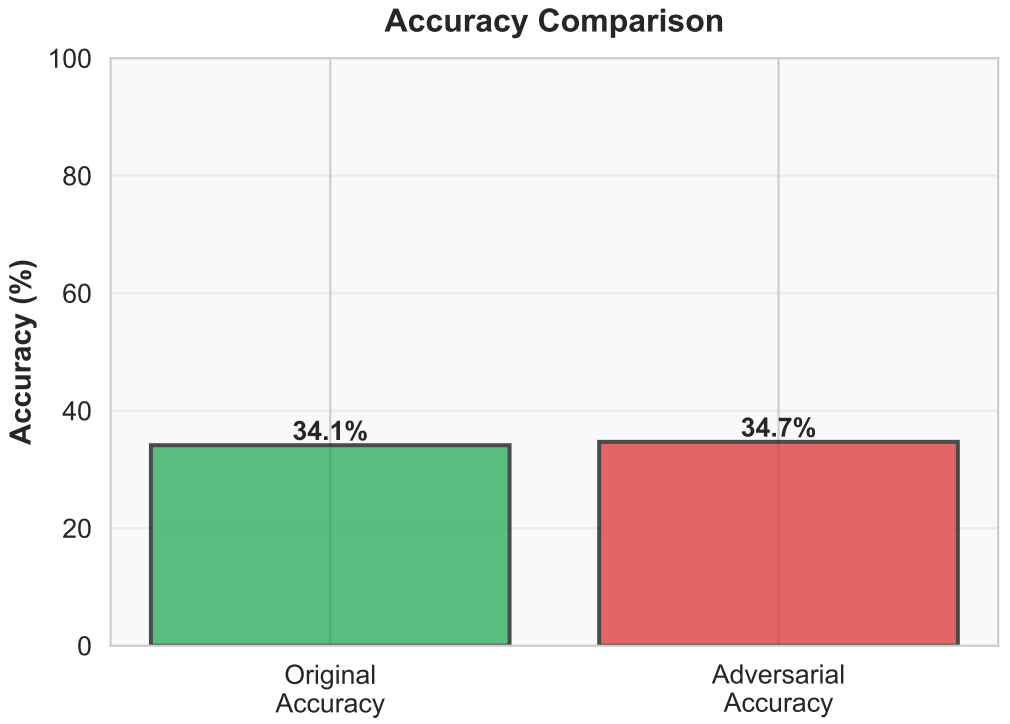
HIGH RISK

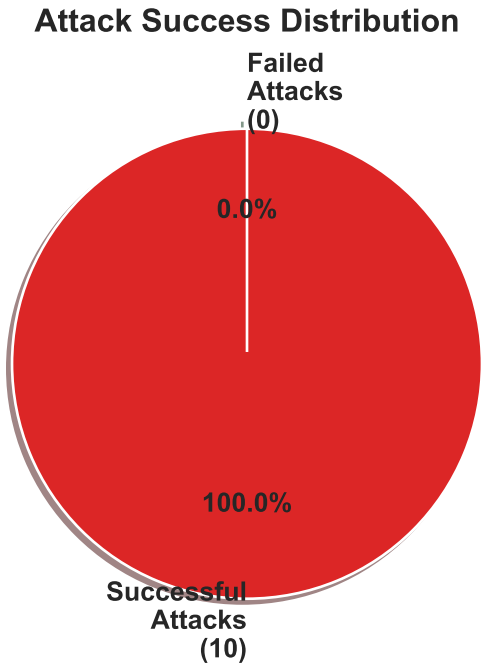
The FGSM attack altered model predictions on 100.0% of the evaluated images. Average confidence shifted from 34.13% to 34.70%, indicating the current robustness posture is high risk.

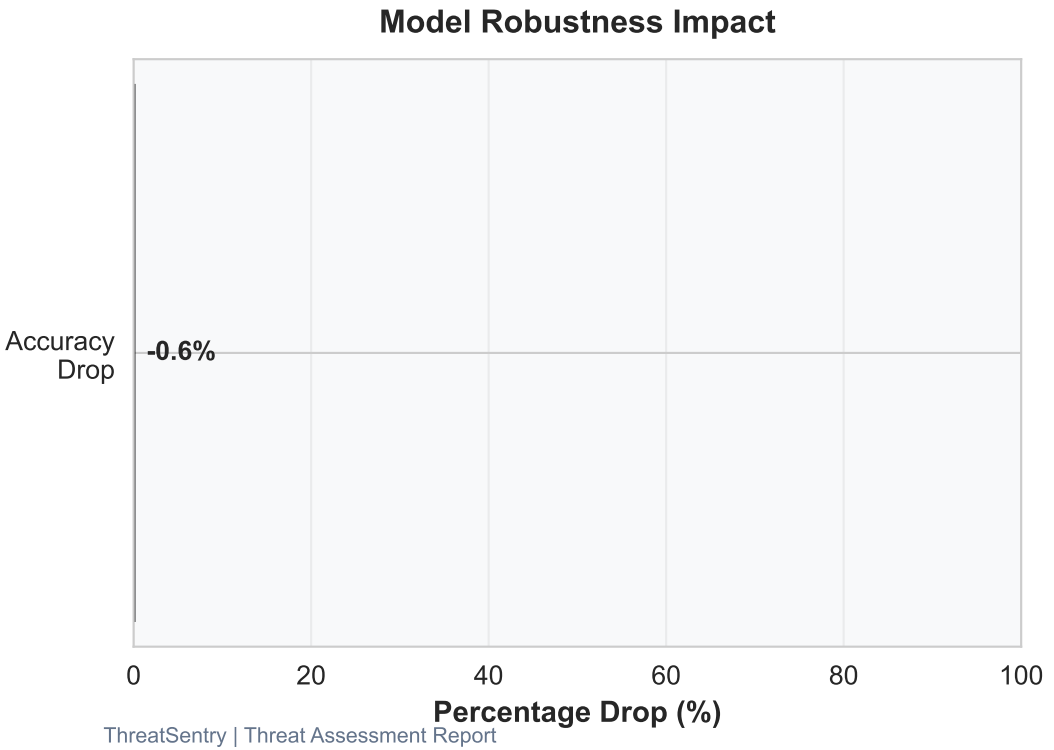
Assessment Narrative

Successfully executed FGSM attack on model 20260206_171419_resnet50_classifier.pt using 10 test images. Attack success rate: 100.0%. Average original accuracy: 34.13%, Average adversarial accuracy: 34.70%. The attack successfully fooled the model in 10 out of 10 cases.

Detailed Analysis







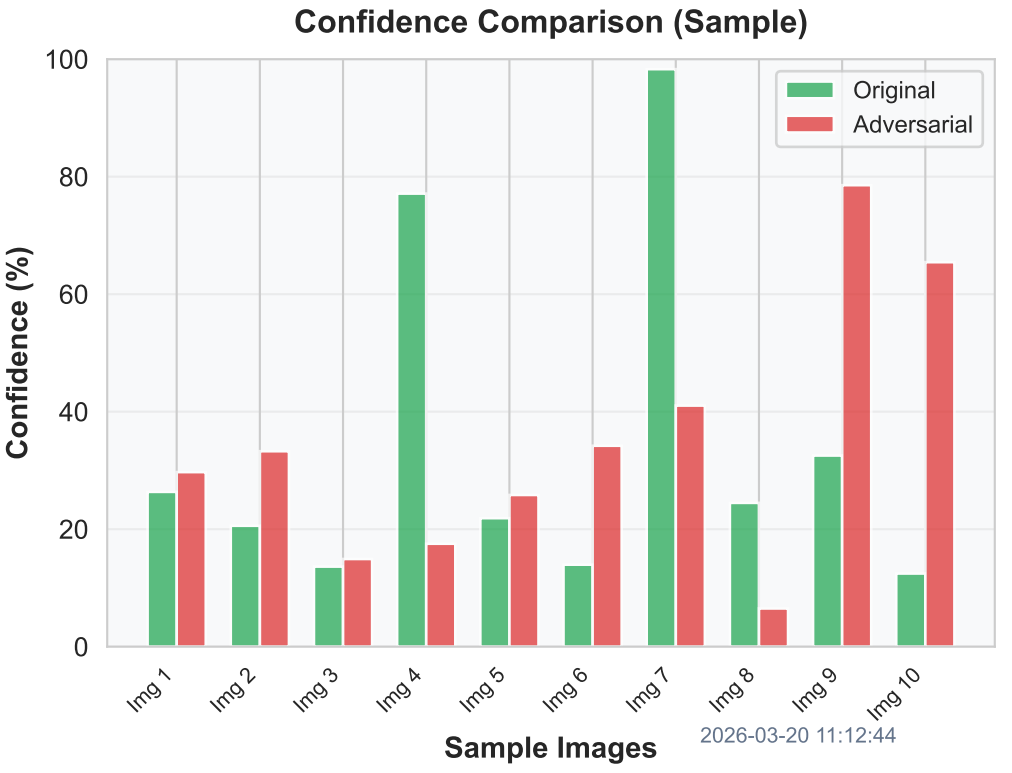


Image Evidence and Predictions

Images 1-4 of 10

Attack Successful
Image: electric_locomotive.JPEG
Outcome: Attack Successful
Model predicted as: Class 730
Original confidence: 12.45%
After attack predicted as: Class 611
Adversarial confidence: 65.45%

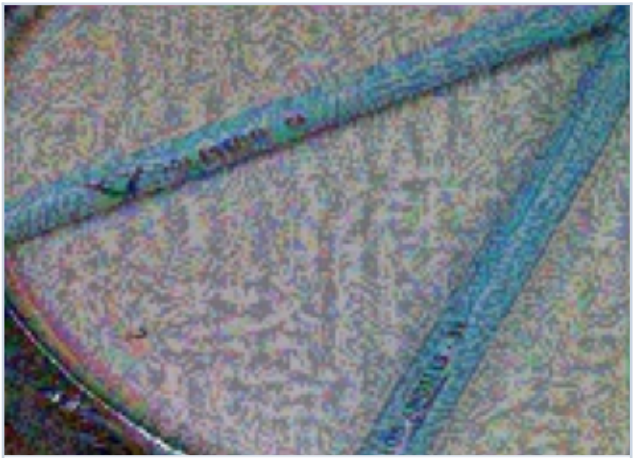
Attack Successful
Image: peacock.JPEG
Outcome: Attack Successful
Model predicted as: Class 84
Original confidence: 77.12%
After attack predicted as: Class 411
Adversarial confidence: 17.52%

Attack Successful
Image: axolotl.JPEG
Outcome: Attack Successful
Model predicted as: Class 973
Original confidence: 21.87%
After attack predicted as: Class 549
Adversarial confidence: 25.82%

Attack Successful
Image: marimba.JPEG
Outcome: Attack Successful
Model predicted as: Class 642
Original confidence: 98.30%
After attack predicted as: Class 611
Adversarial confidence: 41.04%

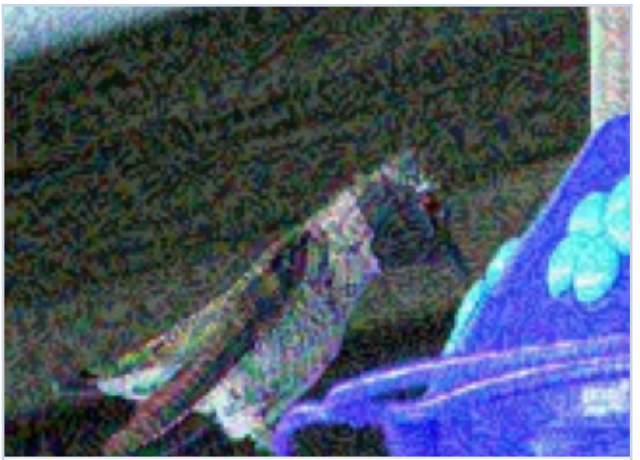
Image Evidence and Predictions

Attack Successful
Image: patas.JPEG
Outcome: Attack Successful
Model predicted as: Class 331
Original confidence: 24.48%
After attack predicted as: Class 549
Adversarial confidence: 6.51%



Attack Successful
Image: drumstick.JPEG
Outcome: Attack Successful
Model predicted as: Class 893
Original confidence: 26.35%
After attack predicted as: Class 669
Adversarial confidence: 29.72%

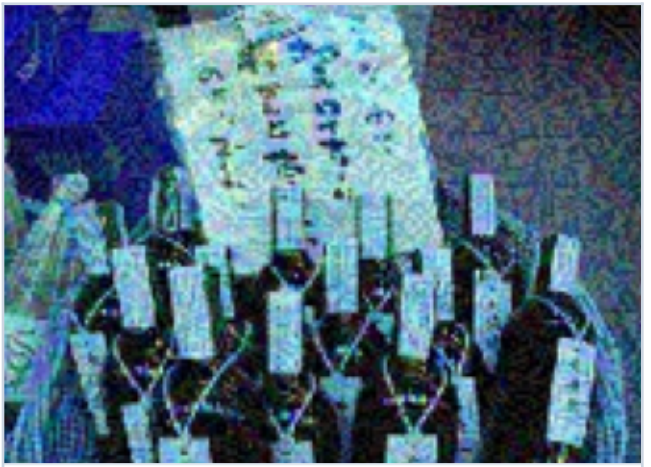
Attack Successful
Image: Border_collie.JPEG
Outcome: Attack Successful
Model predicted as: Class 232
Original confidence: 32.54%
After attack predicted as: Class 611
Adversarial confidence: 78.56%



Attack Successful
Image: hummingbird.JPEG
Outcome: Attack Successful
Model predicted as: Class 94
Original confidence: 13.96%
After attack predicted as: Class 549
Adversarial confidence: 34.20%

Image Evidence and Predictions

Images 9-10 of 10



Attack Successful
Image: wine_bottle.JPEG
Outcome: Attack Successful
Model predicted as: Class 878
Original confidence: 13.63%
After attack predicted as: Class 721
Adversarial confidence: 14.93%

Attack Successful
Image: cuirass.JPEG
Outcome: Attack Successful
Model predicted as: Class 550
Original confidence: 20.59%
After attack predicted as: Class 794
Adversarial confidence: 33.27%

Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed FGSM attack on model 20260206_171419_resnet50_classifier.pt using 10 test images. Attack success rate: 100.0%. Average original accuracy: 34.13%, Average adversarial accuracy: 34.70%. The attack successfully fooled the model in 10 out of 10 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques